

Assignment No -10

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Title - Study and compare AI model deployment techniques across cloud, on-premise, and edge computing environments. Analyze the trade-offs between latency, scalability, and cost.

```
import pandas as pd
import numpy as np
import time
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
df = pd.read_csv("Bank Customer Churn Prediction.csv")
```

df

	customer_id	credit_score	country	gender	age	tenure	balance	products_number	credit_card	active_member	estimated_
0	15634602	619	France	Female	42	2	0.00	1	1	1	10
1	15647311	608	Spain	Female	41	1	83807.86	1	0	1	11
2	15619304	502	France	Female	42	8	159660.80	3	1	0	11
3	15701354	699	France	Female	39	1	0.00	2	0	0	9
4	15737888	850	Spain	Female	43	2	125510.82	1	1	1	7
...	...	...	...	...	...	...	...	...	...	...	...
9995	15606229	771	France	Male	39	5	0.00	2	1	0	9
9996	15569892	516	France	Male	35	10	57369.61	1	1	1	10
9997	15584532	709	France	Female	36	7	0.00	1	0	1	4
9998	15682355	772	Germany	Male	42	3	75075.31	2	1	0	9
9999	15628319	792	France	Female	28	4	130142.79	1	1	0	3

10000 rows × 12 columns

df.describe()

	customer_id	credit_score	age	tenure	balance	products_number	credit_card	active_member	estimated
count	1.000000e+04	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	1.569094e+07	650.528800	38.921800	5.012800	76485.889288	1.530200	0.705500	0.515100	10000.000000
std	7.193619e+04	96.653299	10.487806	2.892174	62397.405202	0.581654	0.455840	0.499797	575.000000
min	1.556570e+07	350.000000	18.000000	0.000000	0.000000	1.000000	0.000000	0.000000	10000.000000
25%	1.562853e+07	584.000000	32.000000	3.000000	0.000000	1.000000	0.000000	0.000000	510.000000
50%	1.569074e+07	652.000000	37.000000	5.000000	97198.540000	1.000000	1.000000	1.000000	10000.000000
75%	1.575323e+07	718.000000	44.000000	7.000000	127644.240000	2.000000	1.000000	1.000000	1493.000000
max	1.581569e+07	850.000000	92.000000	10.000000	250898.090000	4.000000	1.000000	1.000000	1999.000000

```
df.shape
```

(10000, 12)

```
df.isnull().sum()
```

	0
customer_id	0
credit_score	0
country	0
gender	0
age	0
tenure	0
balance	0
products_number	0
credit_card	0
active_member	0
estimated_salary	0
churn	0

dtype: int64

df = df.drop("customer_id", axis=1)

categorical_cols = ["country", "gender"]

```
for col in categorical_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
```

```
X = df.drop("churn", axis=1)
y = df["churn"]
```

```
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = RandomForestClassifier(n_estimators=150, random_state=42)
model.fit(X_train, y_train)
```

▼ RandomForestClassifier ⓘ ?

RandomForestClassifier(n_estimators=150, random_state=42)

y_pred = model.predict(X_test)

```
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report\n")
print(classification_report(y_test, y_pred))
```

Accuracy: 0.864

Classification Report

	precision	recall	f1-score	support
0	0.88	0.96	0.92	1607
1	0.74	0.47	0.57	393
accuracy			0.86	2000
macro avg	0.81	0.71	0.75	2000
weighted avg	0.85	0.86	0.85	2000

```
def on_premise_deployment():
    start = time.time()
```

```

model.predict(X_test)
end = time.time()
return end - start

```

```

def cloud_deployment():

    network_delay = 0.25
    start = time.time()
    time.sleep(network_delay)
    model.predict(X_test)
    end = time.time()
    return end - start

```

```

def edge_deployment():

    edge_data = X_test[:40]
    start = time.time()
    model.predict(edge_data)
    end = time.time()
    return end - start

```

```

on_prem_time = on_premise_deployment()
cloud_time = cloud_deployment()
edge_time = edge_deployment()

```

```

print("\nDeployment Latency Comparison")
print("-----")
print("On-Premise Latency :", round(on_prem_time,4),"seconds")
print("Cloud Latency      :", round(cloud_time,4),"seconds")
print("Edge Latency       :", round(edge_time,4),"seconds")

```

```

Deployment Latency Comparison
-----
On-Premise Latency : 0.0512 seconds
Cloud Latency      : 0.2993 seconds
Edge Latency       : 0.0105 seconds

```

```

print("\nDeployment Trade-Off Analysis")

print("\nCloud:")
print("High scalability, but higher latency due to network and pay-as-you-go cost.")

print("\nOn-Premise:")
print("Low latency with strong security but requires expensive infrastructure.")

print("\nEdge Computing:")
print("Very low latency suitable for real-time applications but limited computing power.")

```

```

Deployment Trade-Off Analysis

Cloud:
High scalability, but higher latency due to network and pay-as-you-go cost.

On-Premise:
Low latency with strong security but requires expensive infrastructure.

Edge Computing:
Very low latency suitable for real-time applications but limited computing power.

```

Start coding or [generate](#) with AI.

